Parallels between Minds and Machines

Cognitive Science 16:185:601
Computer Science 16:198:671
3 credits
Fall 2026

Instructor: Qiong Zhang
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Classroom location: Psych 301, Busch campus
Meeting Times: Thursday 10:20pm to 1:20pm
Instructor Office Hour: Thursday 1:20pm to 1:50pm at Psych 213

Course Description

This interdisciplinary graduate-level course explores the parallels between human learning and machine learning. The central link between the two is the set of shared computational problems faced by humans and machines which includes making complex decisions; predicting future events; storing and retrieving information efficiently; and generalizing knowledge to new situations. By examining such problems, we will see that

1. solutions drawn on methods developed from machine learning can help us gain insights about human cognition, and conversely,
2. knowledge about how humans solve these problems can inform the development of more intelligent machines.

The first half of the course covers the application of machine learning to explain how human cognition works. We will explore the landscape of computational models of human cognition and discuss the insights these models reveal into how people learn, remember, and make complex decisions in everyday situations. The methods discussed include neural networks, symbolic approaches, Bayesian statistics, and more. The applications discussed include perception, skill learning, memory, categorization, and decision making.

In the second half of the course we will draw parallels between human learning and machine learning. Specifically, we will explore how neuroscience and our understanding of human cognition can explain and inform advances in machine learning. We will accomplish this by examining recent advances in neural networks and reinforcement learning from a psychologist's perspective.

Each class will start with a short lecture covering the necessary machine learning techniques and cognitive science concepts to understand the readings. Following this is a student presentation of the reading. We will end with a discussion around the reading.

Learning Objectives

By the end of the course, students will

1. understand the basics of Bayesian inference, neural networks and other computational approaches,
2. understand the basics of the key aspects of human cognition such as memory and decision making,
3. be able to characterize the relationship between computational approaches to cognition and machine learning research
4. be able to identify ways in which computational models can be experimentally tested as models of cognition, and
5. be able to critically discuss the relationship between human intelligence and machine intelligence

Textbook/Resources

Lecture slides are self-contained. There is no required textbook. There will be a number of cognitive science and computer science papers for discussion, available as PDF files through the class website.

Who should take this course

The course is designed for graduate students in cognitive science, psychology, computer science or philosophy who are interested in developing computational models of human cognition and exploring the parallels between human learning and machine learning.

Coursework Requirements

Students are expected to actively participate in class discussions and sign up for at least one class ***presentation*** (20% of total grade).

There will be one to two reading assignments for every class, and you are expected to arrive in class with ideas and questions to discuss. To help you develop these ideas, you are required to write short ***commentaries*** before classes– one to two paragraphs is typical for each paper (20% of total grade). A commentary might take one or several of the following forms: describe the part of the reading that you find most interesting or surprising; mention a claim that doesn't seem right to you; describe how the work could be usefully extended; draw a connection between the reading and something else that has been discussed previously. The one-paragraph commentary should be ended with a suggestion on what would be a good discussion question to have in class. Commentaries are graded pass/fail. If you submit and pass all commentaries, you will receive full credit for this component of the course.

Students are expected to attend all classes and take notes on the most basic and important concepts discussed in each class. There will be a number of in-class ***quizzes*** distributed randomly across the semester. They consist of straightforward short true and false questions which serve as attendance and attention check for that class.

Another component of the course is an individual/team ***project*** to assess the student's ability to put together the concepts and tools they have learned in the course (50% of total grade), delivered by a mid-term report, a final report, and a final presentation. The class project will be an independent research project analyzing an experiment, testing a new cognitive/machine learning model, analyzing an existing model, or synthesizing a controversial topic in the literature. This project will be an excellent opportunity for students to engage in multi-disciplinary research.

Grade Evaluation

Commentaries (due midnight prior to each class)	20%
Paper presentations	20%
In-class quizzes/attendance	10%
Project mid-term report	20%
Project final report	20%
Project final presentation	10%

Schedule of Classes and Readings

Week 1 – Sep 3

Course Overview
Review of key concepts in human cognition; history of cognitive modeling; human intelligence and machine intelligence

Week 2 – Sep 10

Marr's three levels of analysis
Marr, D. (1982). Vision. San Francisco: W. H. Freeman. Chapter 1.

Hamrick, J., & Mohamed, S. (2020). Levels of analysis for machine learning. arXiv preprint arXiv:2004.05107.

Class project briefing
Overview of class projects and datasets

Week 3 – Sep 17

Using cognitive psychology to understand LLMs

Binz, M., & Schulz, E. (2023). Using cognitive psychology to understand GPT-3. Proceedings of the National Academy of Sciences, 120(6),

Using LLMs to understand human mind
*Binz, M., Akata, E., Bethge, M. et al. A foundation model to predict and capture human cognition. Nature **644**, 1002–1009 (2025).*

Week 4 – Sep 24

Formal theories of cognition
van Rooij, I., & Blokpoel, M. (2020). Formalizing verbal theories: A tutorial by dialogue. Social Psychology, 51(5), 285–298. https://doi.org/10.1027/1864-9335/a000428

Mechanistic models of cognition (Sep 20)
Luhmann, C. C., & Rajaram, S. (2015). Memory transmission in small groups and large networks: An agent-based model. Psychological Science, 26(12), 1909-1917.

Week 5 – Oct 1

Rational analysis

Schooler, L. J., & Anderson, J. R. (2017). The Adaptive Nature of Memory. In J. H. Byrne (Ed.) Learning and Memory: A Comprehensive Reference, 2nd edition. Amsterdam, Elsevier. (Originally: Anderson, J. R. (1990). The adaptive character of thought. Hillsdale, NJ: Erlbaum. Chapter 1.)

Griffiths, T. L., Steyvers, M., & Firl, A. (2007). Google and the mind: Predicting fluency with PageRank. Psychological science, 18(12), 1069-1076.

Week 6 – Oct 8

Probabilistic models of cognition

Tenenbaum, J. B. (2000). Rules and similarity in concept learning. Advances in neural information processing systems, 12, 59-65.

Huttenlocher, J., Hedges, L.V., & Vevea, J.L. (2000). Why do categories affect stimulus judgment? Journal of Experimental Psychology, General, 129, 220-241

Week 7 – Oct 15
Resource-rational analysis

Lieder, F., Griffiths, T. L., & Goodman, N. D. (2012, December). "Burn-in, bias, and the rationality of anchoring". In NIPS (pp. 2699-2707).

Ma, S., Popov, V., & Zhang, Q. (2026). Cognitive Constraints and Reward Environments Jointly Shape Memory Formation. Psychological Review.

Week 8 – Oct 22

Neural network models of cognition

Hinton, G. E., Plaut, D. C., & Shallice, T. (1993). Simulating brain damage. Scientific American, 269(4), 76-82.

Lu, Q., Hasson, U., & Norman, K. A. (2021). When to retrieve and encode episodic memories: a neural network model

Week 9 – Oct 29

Human-machine comparison

Cichy, R. M., Khosla, A., Pantazis, D., Torralba, A., & Oliva, A. (2016). Comparison of deep neural networks to spatio-temporal cortical dynamics of human visual object recognition reveals hierarchical correspondence. Scientific reports, 6(1), 27755.

Dapello, J., Marques, T., Schrimpf, M., Geiger, F., Cox, D. D., & DiCarlo, J. J. (2020). Simulating a primary visual cortex at the front of CNNs improves robustness to image perturbations. Advances in Neural Information Processing Systems, 33, 13073-13087.

Week 10 – Nov 5

Inductive bias

Hermann, K., Chen, T., & Kornblith, S. (2020). The origins and prevalence of texture bias in convolutional neural networks. Advances in Neural Information Processing Systems, 33, 19000-19015.

Zhu, J. Q., & Griffiths, T. L. (2024). Eliciting the priors of large language models using iterated in-context learning. Proceedings of the 47th Annual Conference of the Cognitive Science Society

Week 11 – Nov 12

Brain-inspired mechanisms

Van de Ven, G. M., Siegelmann, H. T., & Tolias, A. S. (2020). Brain-inspired replay for continual learning with artificial neural networks. Nature communications, 11(1), 4069.

Konkle, T., & Alvarez, G. A. (2022). A self-supervised domain-general learning framework for human ventral stream representation. Nature communications, 13(1), 491.

Week 12 – Nov 19

Contextual memory (Nov 27)
Ren, M., Iuzzolino, M. L., Mozer, M. C., & Zemel, R. S. (2021). Wandering within a world: Online contextualized few-shot learning. ICLR.

Hierarchical memory (Nov 29)
Lampinen, A. K., Chan, S. C., Banino, A., & Hill, F. (2021). Towards mental time travel: a hierarchical memory for reinforcement learning agents. arXiv preprint arXiv:2105.14039

Week 13 Thanksgiving break

Week 14 – Dec 3

Event understanding in large language models (Dec 3)
Michelmann, S., Kumar, M., Norman, K. A., & Toneva, M. (2023). Large language models can segment narrative events similarly to humans. arXiv preprint arXiv:2301.10297.

Reasoning in large language models (Dec 6)
Sap, M., LeBras, R., Fried, D., & Choi, Y. (2022). Neural theory-of-mind? on the limits of social intelligence in large lms. arXiv preprint arXiv:2210.13312.

Week 15 – Dec 10
Final project presentations

Academic Integrity Policies

Rutgers University regards acts of dishonesty (e.g. plagiarism, cheating on examinations, obtaining unfair advantage, and falsification of records and official documents) as serious offenses against the values of intellectual honesty. Violations of academic integrity will be treated in accordance with university policy, and sanctions for violations may range from no credit for the assignment, to a failing course grade to (for the most severe violations) dismissal from the university. Details policies can be found here: http://academicintegrity.rutgers.edu

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• properly acknowledge all contributors to a given piece of work
• make sure that all work submitted as his or her own in a course or other academic activity is produced without the aid of unsanctioned materials or unsanctioned collaboration

• treat all other students in an ethical manner, respecting their integrity and right to pursue their educational goals without interference. This requires that a student neither facilitate academic dishonesty by others nor obstruct their academic progress (reproduced from: http://academicintegrity.rutgers.edu/academic-integrity-at-rutgers/).

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